

0.1 Product Space

0.1.1 Finite Product Space

Definition 0.1.1. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be Topological Spaces. Define a Topology on $X \times Y$:

$$\mathcal{T}_P \stackrel{\text{def}}{=} \{U \times V \subset X \times Y \mid U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}$$

The Topological Space $(X \times Y, \mathcal{T}_P)$ is called the *Product Space*.

Lemma 0.1.2. Suppose that $(X \times Y, \mathcal{T}_P)$ is a Product Space for X and Y . Then, the Projection map

$$\pi_X : X \times Y \rightarrow X : (x, y) \mapsto x$$

is Continuous Open Onto map.

Proof. Surjection is Clear,

Continuous: For any open $U \in \mathcal{T}_X$, $\pi_X^{-1}[U] = U \times Y \in \mathcal{T}_P$.

Open: For any open $O \times V \in \mathcal{T}_P$, $\pi_X[O \times V] = O \in \mathcal{T}_X$. Thus, it is Open map. □

Theorem 0.1.3. $\mathcal{T}_P$ is the samllest Topology such that the Projection maps are Continuous.

Proof. Suppost that $\mathcal{T}'$ is Topology on $X \times Y$ such that π_X, π_Y are Continuous map.

Then, for any open $U \in \mathcal{T}_X, V \in \mathcal{T}_Y$, $U \times Y = \pi_X^{-1}[U] \in \mathcal{T}'$ and $X \times V = \pi_Y^{-1}[V] \in \mathcal{T}'$

Thus, the finite intersection $U \times Y \cap X \times V = U \times V \in \mathcal{T}'$. That is, $\beta_p \subset \mathcal{T}'$. □

Theorem 0.1.4. Let X , and $\{Y_i\}_{i=1}^n$ are Topological Spaces. For a map

$$f : X \rightarrow \prod_{i=1}^n Y_i : x \mapsto (f_1(x), f_2(x), \dots, f_n(x))$$

The Follwings Are Equivalent:

- a) f is Continuous map.
- b) For all $i = 1, 2, \dots, n$, $f_i : X \rightarrow Y_i$ is Continuous map.

Proof. a) $\implies$ b). Since π_i is Continuous map, $f_i = \pi_i \circ f$ be a Continuous map.

b) $\implies$ a). Let $B = U_1 \times U_2 \times \dots \times U_n \in \mathcal{T}_P$, and $\pi_i : \prod_{i=1}^n Y_i \rightarrow Y_i$ be a Projection for each $i = 1, 2, \dots, n$. Then,

$$\begin{aligned} B &= U_1 \times U_2 \times \dots \times U_n = (U_1 \times X_2 \times X_3 \times \dots \times X_n) \\ &\quad \cap (X_1 \times U_2 \times X_3 \times \dots \times X_n) \\ &\quad \vdots \\ &\quad \cap (X_1 \times X_2 \times X_3 \times \dots \times U_n) = \bigcap_{i=1}^n \pi_i^{-1}[U_i] \end{aligned}$$

Now,

$$f^{-1}[B] = f^{-1} \left[\bigcap_{i=1}^n \pi_i^{-1}[U_i] \right] = \bigcap_{i=1}^n f^{-1} [\pi_i^{-1}[U_i]] = \bigcap_{i=1}^n (\pi_i \circ f)^{-1}[U_i] = \overbrace{\bigcap_{i=1}^n f_i^{-1}[\underline{U_i}]}^{\text{finite intersection}} \underset{f_i \text{ conti}}{\text{open}} \in \mathcal{T}_X$$

□

Theorem 0.1.5. Let $X_1, \dots, X_n$ are Topological Space and $A_i \subset X_i$. Then, for a Product Space $\left(\prod_{i=1}^n X_i, \mathcal{T}_p\right)$,

1. $\overline{\prod_{i=1}^n A_i} = \prod_{i=1}^n \overline{A_i}$.
2. $\left(\prod_{i=1}^n A_i\right)^\circ = \prod_{i=1}^n A_i^\circ$.

Proof.

1.)

$$\begin{aligned}
 x \in \overline{\prod_{i=1}^n A_i} &\iff \forall U \in \mathcal{T}_p \text{ with } x \in U, U \cap \prod_{i=1}^n A_i = \prod_{i=1}^n \pi_i[U] \cap \prod_{i=1}^n A_i = \prod_{i=1}^n (\pi_i[U] \cap A_i) \neq \emptyset \\
 &\iff \forall U \in \mathcal{T}_p \text{ with } x \in U, \forall i = 1, 2, \dots, n, \pi_i[U] \cap A_i \neq \emptyset \\
 &\iff \forall i = 1, 2, \dots, n, \forall U_i \in \mathcal{T}_i \text{ with } \pi_i(x) \in U_i, U_i \cap A_i \neq \emptyset \\
 &\iff x \in \prod_{i=1}^n \overline{A_i}
 \end{aligned}$$

2.)

$$\begin{aligned}
 x \in \left(\prod_{i=1}^n A_i\right)^\circ &\iff \exists U \in \mathcal{T}_p \text{ s.t } x \in U \subset \prod_{i=1}^n A_i \\
 &\iff \exists U \in \mathcal{T}_p \text{ s.t } x \in \prod_{i=1}^n \pi_i[U] \subset \prod_{i=1}^n A_i \\
 &\iff \exists U \in \mathcal{T}_p \text{ s.t } \forall i = 1, 2, \dots, n, \pi_i(x) \in \pi_i[U] \subset A_i \\
 &\stackrel{(*)}{\iff} x \in \prod_{i=1}^n A_i^\circ
 \end{aligned}$$

However, the left direction of $(*)$ fails when the index set is infinite. □